Electric potential is a measure of the work done on unit positive charge to bring it to that point against all electrical forces.

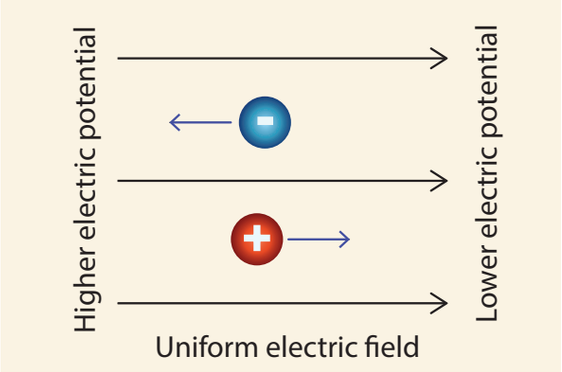


Figure 4.3 Electric potential and Electric field

4.2 Electric current

When the charged object is provided with a conducting path, electrons start to flow through the path from lower potential to higher potential region. Normally, the potential difference is produced by a cell or battery. When the electrons move, we say that an electric current is produced. That is, an electric current is formed by moving electrons.

4.2.1 Direction of current

Before the discovery of the electrons, scientists believed that an electric current consisted of moving positive charges. Although we know this is wrong, the idea is still widely held, as the discovery of the flow of electrons did not affect the basic understanding of the electric current. The movement of the positive charge is called as 'conventional current'. The flow of electrons is termed as 'electron current'. This is depicted in Figure 4.4.

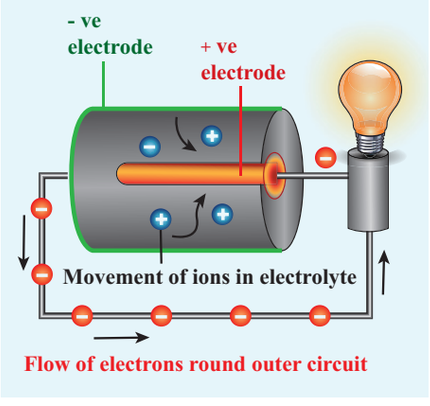


Figure 4.4 Electric current

In electrical circuits the positive terminal is represented by a long line and negative terminal as a short line. Battery is the combination of more than one cell (Fig. 4.5).

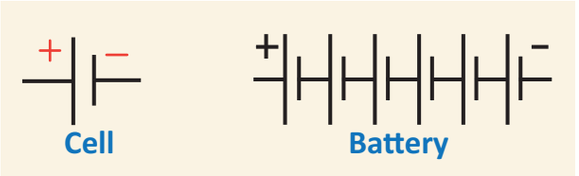


Figure 4.5 Cell and battery

4.2.2 Measurement of electric current

We can measure the value of current and express it numerically. **Current is the rate at which charges flow past a point on a circuit.** That is, if q is the quantity of charge passing through a cross section of a wire in time t, quantity of current (I) is represented as,

$$I = q/t$$

The standard SI unit for current is ampere with the symbol A. Current of 1 ampere means that there is one coulomb (1C) of charge passing through a cross section of a wire every one second (1 s).

1 ampere = 1 coulomb / 1 second (or)
1 A = 1 C / 1 s = 1Cs^{-1}

Ammeter is an instrument used to measure the strength of the electric current in an electric circuit.

The ammeter is connected in series in a circuit where the current is to be found. . The current flows through the positive (+) red terminal of ammeter and leaves from the negative (–) black terminal.

IX_SCI_EM_Unit-04_PHY.indd 41 12/7/2022 5:01:31 PM